

Janit Rajkarnikar

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Research Interests

Machine learning, representation learning, and information access. Interested in privacy, interpretability, and building deployable systems that support trustworthy decision-making. Open to exploring new directions across ML and information science.

Education

University of Southern Mississippi

May 2026

Bachelor of Science in Computer Science, Minor in Mathematics and Data Analysis

Hattiesburg, MS

GPA: 4.0

Coursework: Algorithms, Machine Learning, Data Structures, Operating Systems, Real Analysis, Linear Algebra, Discrete Mathematics, Probability & Statistics, Database Systems, Programming Languages

Honors & Awards

President's List (All Semesters), Academic Excellence Scholarship, Runner-up Voxo Hackathon at USM

Publications

- [1] **J. Rajkarnikar**, N. Poudel, and N. Rahimi, "Unsupervised Anomaly Detection in OpenStack Logs via Fine-Tuned RoBERTa Embeddings," *Journal of Cybersecurity, Digital Forensics and Jurisprudence*, vol. 1, pp. 9–21, 2025. [\[Paper\]](#)
- [2] D. Precious-Esue, **J. Rajkarnikar**, B. Bellrose, et al., "Ensemble Machine Learning Approach to Phishing Website Detection," in *Proc. 40th International Conference on Computers and Their Applications (CATA 2025)*, San Francisco, CA, USA, Mar. 2025, pp. 109–121. Springer CCIS, vol. 2435. [\[DOI\]](#)

Under Preparation:

J. Rajkarnikar, S. Joshi, and Z. Zhou, "Cross-Attention Macro-Market Fusion for Market Movement Classification."

Research Experience

Advised by Dr. Zhaoxian Zhou

Sep 2025 – Present

- Developed **CAMMF**, a cross-attention framework for integrating heterogeneous time-series information from macroeconomic indicators and market signals for return prediction
- Designed alignment and leakage controls for multi-source temporal data; improved AUROC by 8.9% over baselines
- Built interpretability analyses that attribute predictions to specific information sources/lags across regimes (attention attribution + **SHAP**)

Advised by Dr. Nick Rahimi

May 2024 – Present

- Led representation learning pipeline for unstructured OpenStack logs using **RoBERTa** embeddings with **LoRA/PEFT**; achieved **0.97 F1** and **0.99 ROC-AUC** with Isolation Forest (*Publication [1]*)
- Conducted empirical study of ensemble learning for phishing website detection; Bagging achieved **98.66% accuracy** (*Publication [2]*)

Projects

RepoSynth – Repository Understanding & Retrieval Platform (reposynth.com) *Python, Rust, TypeScript*

- Built pipeline for extracting semantic representations and structural signals (AST/dependency graphs) to support repository understanding and retrieval (**SentenceTransformers** + **FAISS**)
- Implemented **Rust Tree-sitter** parsing daemon and multi-stage caching, reducing re-analysis time by 95%

Emoki – Emotion Inference System

React Native, Android, Flask, ONNX

- Deployed quantized **RoBERTa (ONNX)** for real-time affect inference; engineered end-to-end client/server pipeline
- Built Android Accessibility-based text capture and on-device processing workflow, highlighting system design tradeoffs between inference utility and user privacy

Suga – Workflow Automation Companion (suga.cx)

FastAPI, Next.js, Supabase, Electron

- Built an automation capture-and-replay system that converts observed user actions into reusable workflows

Experience

Software Intern

Sep 2025 – Present

CloudCrust LLC

Hattiesburg, MS

- Built and shipped production web/mobile features (**Next.js, React Native**) including geospatial visualization and location-based data workflows
- Implemented authentication and subscription flows for deployed applications

Software Engineer Intern

Aug 2024 – May 2025

Woafmeow, Inc.

Remote

- Developed backend features in **FastAPI/PostgreSQL** supporting secure multi-tenant workflows

Technical Skills

ML/Data: PyTorch, SentenceTransformers, FAISS, SHAP, ONNX, SQL, PostgreSQL, Redis, MongoDB

Systems/Backend: FastAPI, Flask, Node.js/Express, Docker, AWS, Azure, Supabase, Git

Languages: Python, TypeScript, C/C++, C#

Frontend/Mobile: React, Next.js, React Native, Expo, Electron

Service & Leadership

Vice Head of SWE, Google Developer Group – Coordinated technical workshops and campus engineering events

President, Nepalese Student Association – Led 15+ volunteers; organized 250+ attendee cultural/academic events

Treasurer, Cinema Appreciation Club – Managed finances and coordinated screenings/events